True or false? (unless otherwise specified.)

Answer grid

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	B
True	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
False	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

1. **Degree of precision.** The degree of precision of the following quadrature rule is 3:
$$\int_{-1}^{1} f(x)\,\mathrm{d}x \approx f\left(-\frac{1}{3}\right) + f\left(\frac{1}{3}\right). \tag{1}$$

2. **Degree of precision.** The degree of precision of the following quadrature rule is 1:
$$\int_{-1}^{1} f(x)\,\mathrm{d}x \approx 2f(0) - f(-1) + f(1).$$

3. **Degree of precision.** The degree of precision of the following rule is 5:
$$\int_{-1}^{1} f(x)\,\mathrm{d}x \approx 2f(0) + \frac{1}{3}f''(0) + \frac{1}{60}f^{(4)}(0).$$

4. **Degree of precision.** The degree of precision of the following quadrature rule is 3:
$$\int_{-1}^{1} f(x)\,\mathrm{d}x \approx \frac{4}{3}f\left(-\frac{1}{2}\right) - \frac{2}{3}f(0) + \frac{4}{3}f\left(\frac{1}{2}\right).$$

5. **Degree of precision.** For any natural number $N > 0$ and distinct nodes $x_0 < \ldots < x_N$, there exist weights $w_0, \ldots, w_N$ such that the quadrature rule
$$\int_{-1}^{1} u(x)\,\mathrm{d}x \approx \sum_{n=0}^{N} w_n u(x_n)$$
has a degree of precision at least equal to N.

6. **Degree of precision.** The closed Newton–Cotes rule with 3 equally spaced points (Simpson's rule) is exact for all polynomials of degree up to 3.

7. **Composite quadrature.** Simpson's composite integration rule is exact for every polynomial f, whatever its degree.

8. **Composite quadrature.** With $h = \frac{b-a}{n}$ and $x_i = a + ih$, the following formula is the composite version, with n cells, of the quadrature rule (1):
$$\int_a^b f(x)\,\mathrm{d}x \approx \frac{h}{2}\sum_{i=0}^{n-1}\left(f\left(x_i + \frac{h}{3}\right) + f\left(x_i + \frac{2h}{3}\right)\right). \tag{2}$$

9. **Convergence estimate.** Suppose that $f \in C^\infty[a,b]$ and let $I_n[f]$ denote the approximate integral of f using the composite trapezoidal rule with $n+1$ equidistant integration points. Then there exists $C \in (0,\infty)$ such that
$$\forall n \in \{1,2,\ldots\}, \qquad \Big|I[f] - I_n[f]\Big| \leqslant \frac{C}{n^2}, \qquad I[f] := \int_a^b f(x)\,\mathrm{d}x.$$

10. **Convergence estimate.** With the notation of the previous item, it holds for every $f \in C^\infty[a,b]$ that
$$\lim_{n\to\infty} n^2\Big|I[f] - I_n[f]\Big| = 0.$$

11. **Convergence estimate.** Suppose that $f \in C^\infty[a,b]$ and let $I_n[f]$ denote the approximate integral of f using the composite Simpson rule with $n+1$ equidistant integration points, n even. Then it holds that
$$\lim_{n\to\infty} n^2\Big|I[f] - I_n[f]\Big| = 0.$$

12. **Convergence estimate.** Let $u(x) = (1 + 25x^2)^{-1}$ and let

$$I_N^T = \frac{1}{2N}\Big(u(x_0) + 2u(x_1) + \cdots + 2u(x_{N-1}) + u(x_N)\Big)\,, \qquad x_i = \frac{i}{N}\,,$$

be the composite trapezoidal approximation of $\int_0^1 u(x)\,\mathrm{d}x$. Then I_N^T diverges in the limit $N \to \infty$.

13. **Convergence estimate.** Let $I_n[f]$ denote the right-hand side of the composite formula (2), with n cells. Then for every $f \in C^\infty[a, b]$ it holds that

$$\lim_{n\to\infty} n^3 \Big| I[f] - I_n[f] \Big| = 0, \qquad I[f] := \int_a^b f(x)\,\mathrm{d}x\,.$$

14. **Julia.** The following code implements the composite version, with n cells, of the quadrature rule (1):

```
function I_approx(f, a, b, n)
    x = LinRange(a, b, n + 1)
    h = (b - a) / n
    m = x[1:n] .+ h/2
    return h/2 * sum(f, m .- h/6) + h/2 * sum(f, m .+ h/6)
end
```

15. **Julia.** The following code correctly implements the composite Simpson rule with n cells, where n is even:

```
function I_approx(f, a, b, n)
    x = LinRange(a, b, n + 1)
    h = (b - a) / n
    return h/3 * (f(x[1]) + 2sum(f, x[2:2:end-1])
                  + 4sum(f, x[3:2:end-2]) + f(x[end]))
end
```

16. **Bonus.** There exist nodes x_1, x_2, x_3 and weights w_1, w_2, w_3 such that the rule

$$\int_{-1}^1 u(x)\,\mathrm{d}x \approx w_1 u(x_1) + w_2 u(x_2) + w_3 u(x_3)$$

has a degree of precision equal to 12. *Justify briefly.*

Answer key

Questions 1–15: 1 point for a correct answer, 0 otherwise.

1. **False.** The rule is exact for 1, x and x^3 by symmetry, but for x^2 it gives $\frac{2}{9} \neq \frac{2}{3}$. Its degree of precision is 1.

2. **False.** For $f(x) = x$ the rule gives $2 \cdot 0 + 1 + 1 = 2$, whereas the integral is 0. The degree of precision is 0.

3. **True.** By symmetry the rule is exact for odd monomials. It is exact for 1 (2), for x^2 ($\frac{1}{3} \cdot 2 = \frac{2}{3}$) and for x^4 ($\frac{1}{60} \cdot 24 = \frac{2}{5}$). It is not exact for x^6, for which it gives 0 instead of $\frac{2}{7}$, so the degree of precision is exactly 5.

4. **True.** The rule is exact for 1 ($\frac{4}{3} - \frac{2}{3} + \frac{4}{3} = 2$) and for x^2 ($\frac{4}{3} \cdot \frac{1}{4} \cdot 2 = \frac{2}{3}$), and by symmetry for x and x^3. It is not exact for x^4: it gives $\frac{1}{6}$ instead of $\frac{2}{5}$.

5. **True.** Take $w_n = \int_{-1}^{1} \ell_n(x)\,\mathrm{d}x$, where ℓ_n are the Lagrange polynomials. The rule then integrates exactly the interpolant of u, which is equal to u for every $u \in \mathcal{P}_N$.

6. **True.** Simpson's rule is exact for quadratics by construction and, by symmetry, also for x^3.

7. **False.** The composite rule is exact only for polynomials of degree at most 3. For instance, it is not exact for x^4.

8. **True.** Mapping $[-1, 1]$ to the cell $[x_i, x_{i+1}]$ sends the nodes $\pm\frac{1}{3}$ to $x_i + \frac{h}{2} \pm \frac{h}{6}$, that is, to $x_i + \frac{h}{3}$ and $x_i + \frac{2h}{3}$, and the weights 1 become $\frac{h}{2}$.

9. **True.** The error is bounded by $\frac{(b-a)^3}{12} \frac{\max|f''|}{n^2}$.

10. **False.** The order 2 is sharp in general. For $f(x) = x^2$ on $[0, 1]$ the error is exactly $\frac{1}{6n^2}$, so $n^2|I[f] - I_n[f]| = \frac{1}{6}$ for all n.

11. **True.** For $f \in C^4$ the error of the composite Simpson rule is $O(n^{-4})$, so $n^2|I[f] - I_n[f]| = O(n^{-2}) \to 0$.

12. **False.** The function is continuous, hence Riemann integrable, and I_N^T is a Riemann sum that converges to the integral.

13. **False.** The rule has degree of precision 1, so the error is of order $h^2 = O(n^{-2})$, and this is sharp. For $f(x) = x^2$ on $[0, 1]$, each cell contributes an error $\frac{h^3}{18}$, so $|I[f] - I_n[f]| = \frac{1}{18n^2}$ and $n^3|I[f] - I_n[f]| = \frac{n}{18} \to \infty$.

14. **True.** On the cell with midpoint m and width h, the change of variable $x = m + \frac{h}{2}t$ maps the rule on $[-1, 1]$ to $\frac{h}{2}\big(f(m - \frac{h}{6}) + f(m + \frac{h}{6})\big)$. The code sums this over the n cells, with midpoints `m`.

15. **False.** The weights are swapped. In Julia, `x[2:2:end-1]` are the nodes $x_1, x_3, \ldots, x_{n-1}$ (in the 0-based numbering of the notes), which must get weight $\frac{4h}{3}$, whereas `x[3:2:end-2]` are $x_2, x_4, \ldots, x_{n-2}$, which must get weight $\frac{2h}{3}$. Here it is the opposite.

Bonus. (**False.**) Suppose the degree of precision were 12 (in particular at least 6) and set $q(x) = (x - x_1)^2(x - x_2)^2(x - x_3)^2 \in \mathcal{P}_6$. Then the rule gives 0 while $\int_{-1}^{1} q\,\mathrm{d}x > 0$, a contradiction. In general, n nodes yield a degree of precision at most $2n - 1$, here 5.